Numerical Methods - W3

Evaluating functions by continued fractions

Solving transcendental equations by Rolle's sequence method

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Numerical Methods · Week3 · theory, algorithms and 10 fully worked examples

Roadmap of the presentation

I Evaluating functions by continued fractions

- Definition, notation, convergents
- Backward (bottom-up) evaluation
- Forward recurrence (Wallis–Euler)
- Why CF beat Taylor series
- 5 worked examples: e^x, tan x, ln(1+x), arctan x, √10

II Transcendental equations: Rolle's sequence method

- Rolle's theorem and its corollary
- Building Rolle's sequence, counting sign changes
- Separating every real root
- Refining each root by bisection
- 5 worked examples incl. a two-level Rolle sequence

Part I · Evaluating functions by continued fractions

Replace an infinite expansion by a short chain of divisions — and get many correct digits fast

What is a continued fraction?

General (staircase) form

$$f(x) = b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cfrac{a_3}{b_3 + \cfrac{a_4}{b_4 + \cdots}}}}$$

Compact notation: $f = b_0 + a_1/(b_1+)\ a_2/(b_2+)\ a_3/(b_3+)\ \cdots$

$\mathbf{a_k}$	partial numerators (may depend on x)
$\mathbf{b_k}$	partial denominators (may depend on x)
$\mathbf{C_n}$	n-th convergent: cut the fraction after a_n/b_n
$\mathbf{C_n \to f}$	convergence: the approximation improves with n

Idea: evaluating f(x) = computing a convergent $C_n(x)$, a rational function obtained with only n divisions.

Two ways to evaluate a truncated continued fraction

A Backward (bottom-up) scheme

$$t_n = b_n$$

$$t_k = b_k + a_{k+1} / t_{k+1}$$

$$k = n-1, n-2, \ldots, 1$$

$$f \approx b_0 + a_1 / t_1$$

✓ only n divisions, numerically stable
✗ n must be fixed before starting

B Forward recurrence (Wallis–Euler)

$$A_{-1} = 1,\ A_0 = b_0,\ \ B_{-1} = 0,\ B_0 = 1$$

$$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$$

$$B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$$

$$C_k = A_k / B_k$$

✓ gives every convergent $C_1, C_2, \ldots$ in one sweep
✓ stop when $|C_k - C_{k-1}| < \varepsilon$

Why continued fractions? Comparison with Taylor series

Task: compute $e = e^1$ with comparable work (5 levels / terms up to x^5)

Taylor series, degree 5

2.7166666667

$\mathbf{1.62\times10^{-3}}$

absolute error

Continued fraction, 5 levels

2.7182818287

$\mathbf{2.77\times10^{-10}}$

absolute error

- The CF error is about 6 million times smaller for the same effort.
- CFs approximate f by rational functions (Padé-type), which follow poles and asymptotes better than polynomials.
- CFs often converge on a larger domain: ln(1+x) and arctan x converge for every admissible real x, Taylor only for $|x| \leq 1$.
- Used inside calculators and math libraries for e^x, tan x, erf, Γ, Bessel functions.

e = 2.7182818285 (exact to 10 digits)
Taylor: 1 + 1 + 1/2 + 1/6 + 1/24 + 1/120
CF: 1 + 2/(1 + 1/(6 + 1/(10 + 1/(14 + 1/18))))

Classical expansions used in the five examples

#	Function	Continued fraction expansion	Valid for
1	$\mathbf{e^x}$	$1 + 2x/(2 - x + x^2/(6 + x^2/(10 + x^2/(14 + \cdots))))$	all real x
2	**tan x**	$x/(1 - x^2/(3 - x^2/(5 - x^2/(7 - \cdots))))$ (Lambert)	$x \neq \pi/2 + k\pi$
3	**ln(1 + x)**	$x/(1 + 1^2x/(2 + 1^2x/(3 + 2^2x/(4 + 2^2x/(5 + \cdots)))))$	$x > -1$
4	**arctan x**	$x/(1 + 1^2x^2/(3 + 2^2x^2/(5 + 3^2x^2/(7 + \cdots))))$	all real x
5	$\mathbf{\sqrt{(a^2 + b)}}$	$a + b/(2a + b/(2a + b/(2a + \cdots)))$	$a > 0, b > 0$

Every example follows the same 4 steps: ① read a_k, b_k → ② substitute x → ③ backward evaluation → ④ convergents & error check

CF Example 1 · e^x at x = 1: set up the coefficients

Goal: evaluate $e = e^1$ with n = 5 levels

$$e^x = 1 + \cfrac{2x}{2 - x + \cfrac{x^2}{6 + \cfrac{x^2}{10 + \cfrac{x^2}{14 + \cdots}}}}$$

k	0	1	2	3	4	5
b_k	1	1	6	10	14	18
a_k	—	2	1	1	1	1

1 **Read the pattern of a_k and b_k**

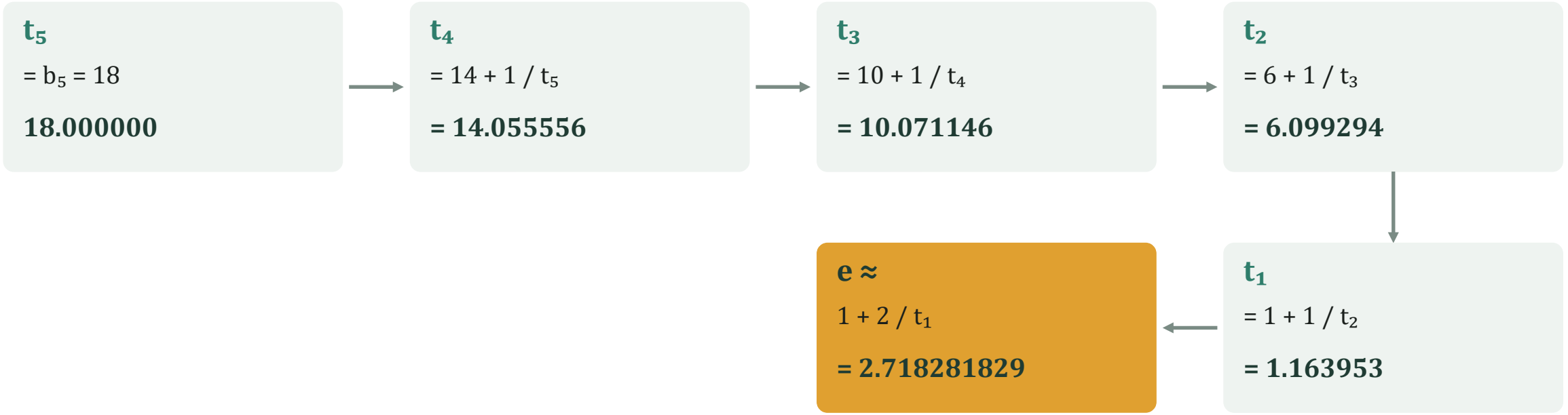
$b_0 = 1$, $a_1 = 2x$, $b_1 = 2 - x$
$a_k = x^2$ $(k \geq 2)$, $b_k = 4k - 2$ $(k \geq 2)$

2 **Substitute the numerical data**

$x = 1 \Rightarrow a_1 = 2, b_1 = 1, a_k = 1, b_k = 6, 10, 14, 18$

Truncation: keep 5 levels, i.e. compute the convergent C_5. The table (left, bottom) is the complete input of both algorithms.

CF Example 1 · Step 3: backward evaluation (bottom-up)

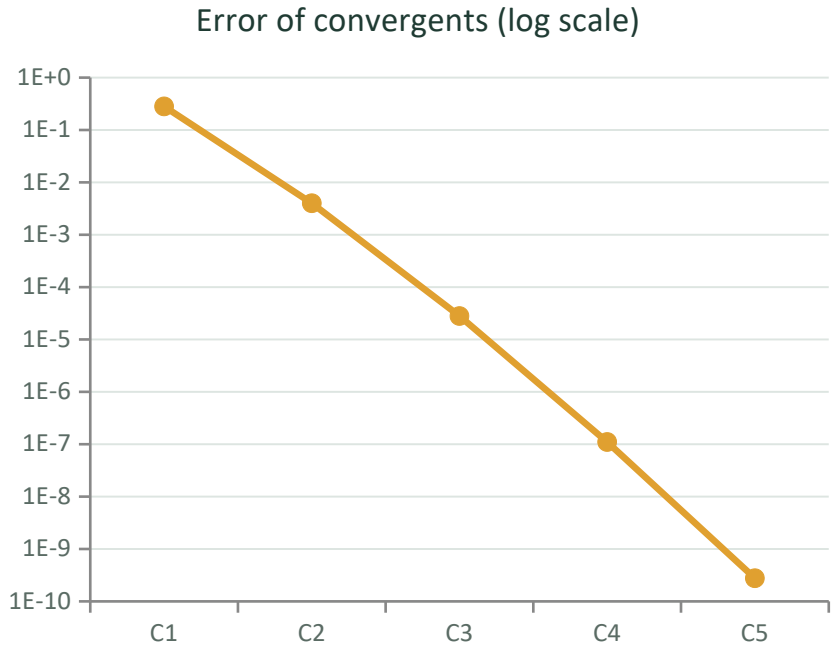


Result: **e ≈ 2.718281829** Exact: e = 2.718281828459…
Absolute error: 2.77×10^{-10} obtained with only 5 divisions. Read the snake in the arrow direction: start at the deepest level t_5.

CF Example 1 · Step 4: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	1	1	1	1.000000000	1.7183
1	2	1	3	1	3.000000000	0.2817
2	1	6	19	7	2.714285714	4.00×10^{-3}
3	1	10	193	71	2.718309859	2.80×10^{-5}
4	1	14	2721	1001	2.718281718	1.10×10^{-7}
5	1	18	49171	18089	2.718281829	2.77×10^{-10}



$C_5 = 2.718281829 \equiv$ backward result ✓
≈ 9 correct decimals

CF Example 2 · tan x at x = 1: set up the coefficients

Goal: evaluate tan 1 with n = 5 levels

$$\tan x = \cfrac{x}{1 - \cfrac{x^2}{3 - \cfrac{x^2}{5 - \cfrac{x^2}{7 - \cdots}}}}$$

k	0	1	2	3	4	5
b_k	0	1	3	5	7	9
a_k	—	1	−1	−1	−1	−1

1 **Read the pattern of a_k and b_k**

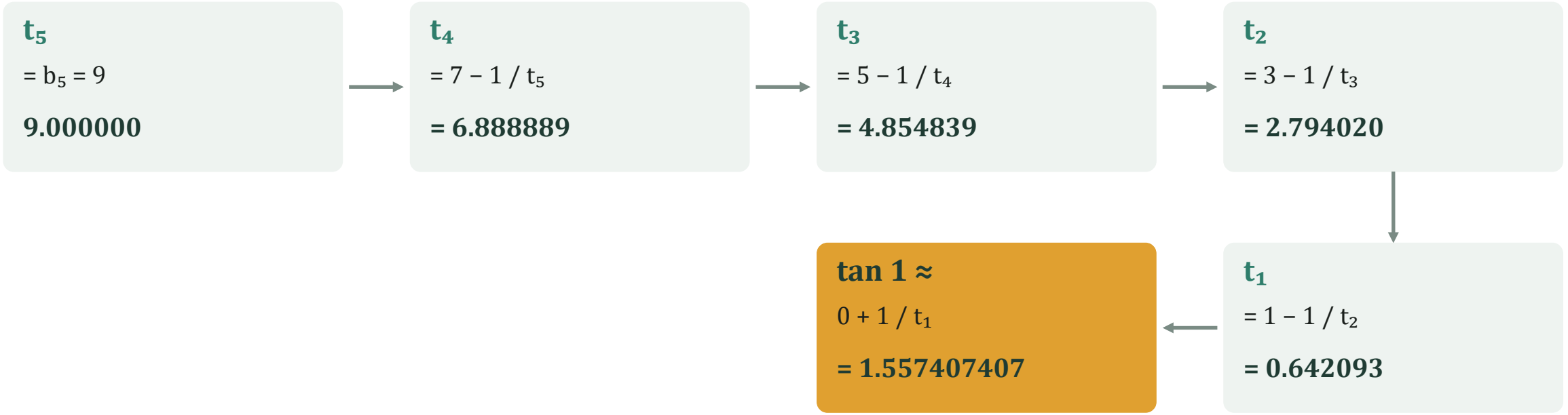
$b_0 = 0$, $a_1 = x$, $a_k = -x^2$ $(k \geq 2)$
$b_k = 2k - 1$ (odd numbers)

2 **Substitute the numerical data**

$x = 1$ (rad) $\Rightarrow$ $a_1 = 1$, $a_k = -1$, $b_k = 1, 3, 5, 7, 9$

Truncation: keep 5 levels, i.e. compute the convergent C_5. The table (left, bottom) is the complete input of both algorithms.

CF Example 2 · Step 3: backward evaluation (bottom-up)

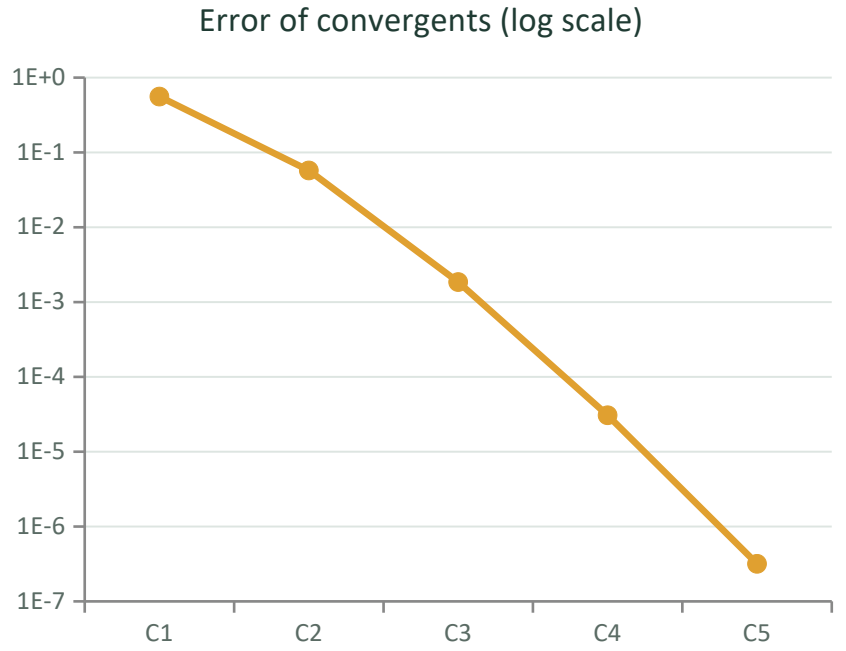


Result: $\tan 1 \approx 1.557407407$ Exact: $\tan 1 = 1.557407724654\ldots$
Absolute error: 3.17×10^{-7} obtained with only 5 divisions. Read the snake in the arrow direction: start at the deepest level t_5.

CF Example 2 · Step 4: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	1.5574
1	1	1	1	1	1.000000000	0.5574
2	−1	3	3	2	1.500000000	0.0574
3	−1	5	14	9	1.555555556	1.85×10^{-3}
4	−1	7	95	61	1.557377049	3.07×10^{-5}
5	−1	9	841	540	1.557407407	3.17×10^{-7}



$C_5 = 1.557407407 \equiv$ backward result ✓
≈ 6 correct decimals

CF Example 3 · ln(1 + x) at x = 1: set up the coefficients

Goal: evaluate ln 2 with n = 7 levels

$$\ln(1+x) = \cfrac{x}{1 + \cfrac{1^2 \cdot x}{2 + \cfrac{1^2 \cdot x}{3 + \cfrac{2^2 \cdot x}{4 + \cdots}}}}$$

k	0	1	2	3	4	5	6	7
b_k	0	1	2	3	4	5	6	7
a_k	—	1	1	1	4	4	9	9

1 Read the pattern of a_k and b_k

$b_0 = 0,\ b_k = k,\ a_1 = x$

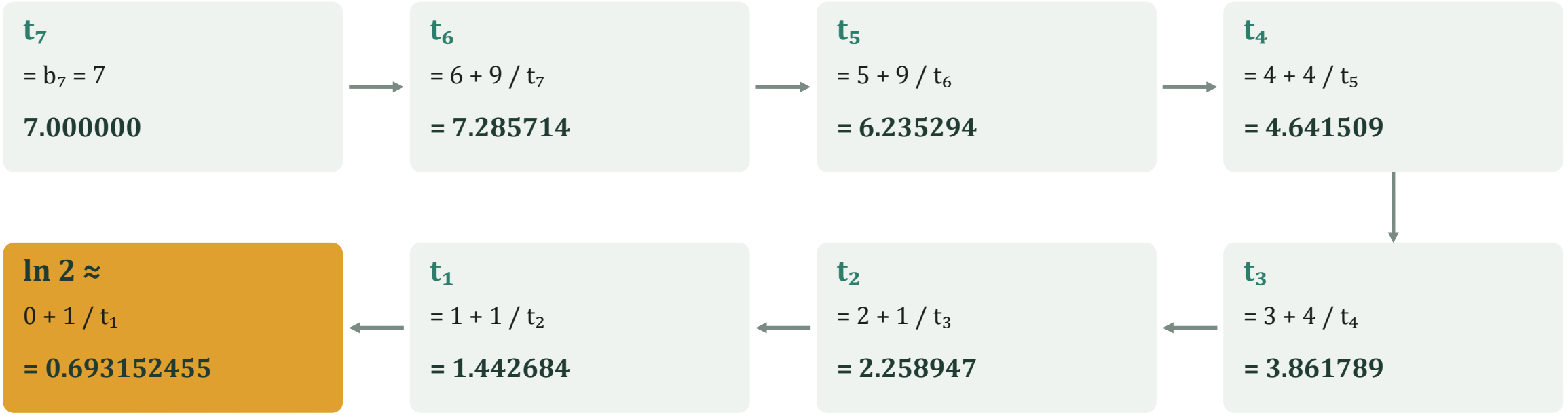
$a_{2m} = a_{2m+1} = m^2 \cdot x \quad (m \geq 1)$

2 Substitute the numerical data

$x = 1 \Rightarrow a_k = 1, 1, 1, 4, 4, 9, 9;\ b_k = 1, 2, \ldots, 7$

Truncation: keep 7 levels, i.e. compute the convergent C_7. The table (left, bottom) is the complete input of both algorithms.

CF Example 3 · Step 3: backward evaluation (bottom-up)

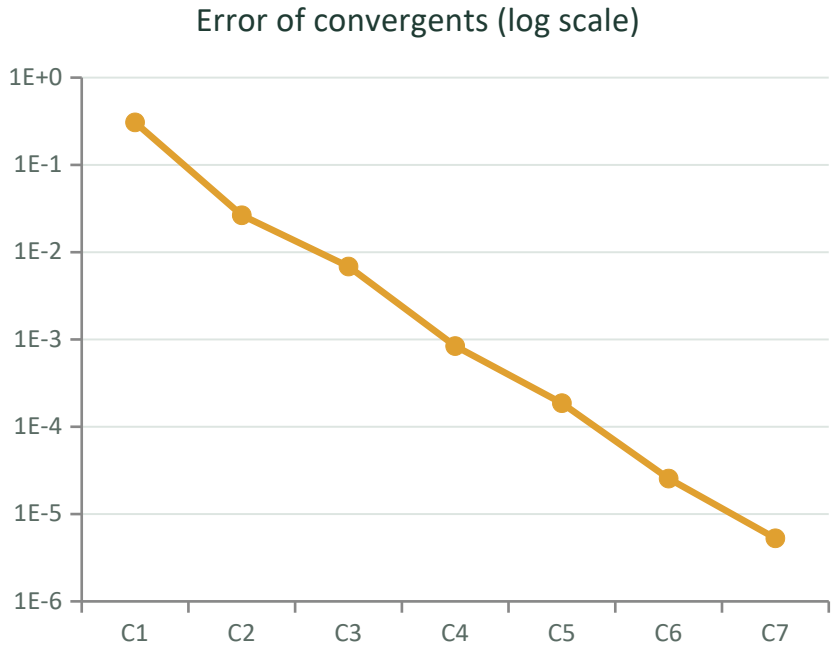


Result: $\ln 2 \approx 0.693152455$ Exact: $\ln 2 = 0.693147180559\ldots$
Absolute error: 5.27×10^{-6} obtained with only 7 divisions. Read the snake in the arrow direction: start at the deepest level t_7.

CF Example 3 · Step 4: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	0.6931
1	1	1	1	1	1.000000000	0.3069
2	1	2	2	3	0.666666667	0.0265
3	1	3	7	10	0.700000000	6.85×10^{-3}
4	4	4	36	52	0.692307692	8.39×10^{-4}
5	4	5	208	300	0.693333333	1.86×10^{-4}
6	9	6	1572	2268	0.693121693	2.55×10^{-5}
7	9	7	12876	18576	0.693152455	5.27×10^{-6}



$C_7 = 0.693152455 \equiv$ backward result ✓
≈ 5 correct decimals

CF Example 4 · arctan x at x = 1: set up the coefficients

Goal: evaluate arctan 1 = π/4 with n = 6 levels

$$\arctan x = \cfrac{x}{1 + \cfrac{1^2x^2}{3 + \cfrac{2^2x^2}{5 + \cfrac{3^2x^2}{7 + \cdots}}}}$$

k	0	1	2	3	4	5	6
b_k	0	1	3	5	7	9	11
a_k	—	1	1	4	9	16	25

1 **Read the pattern of a_k and b_k**

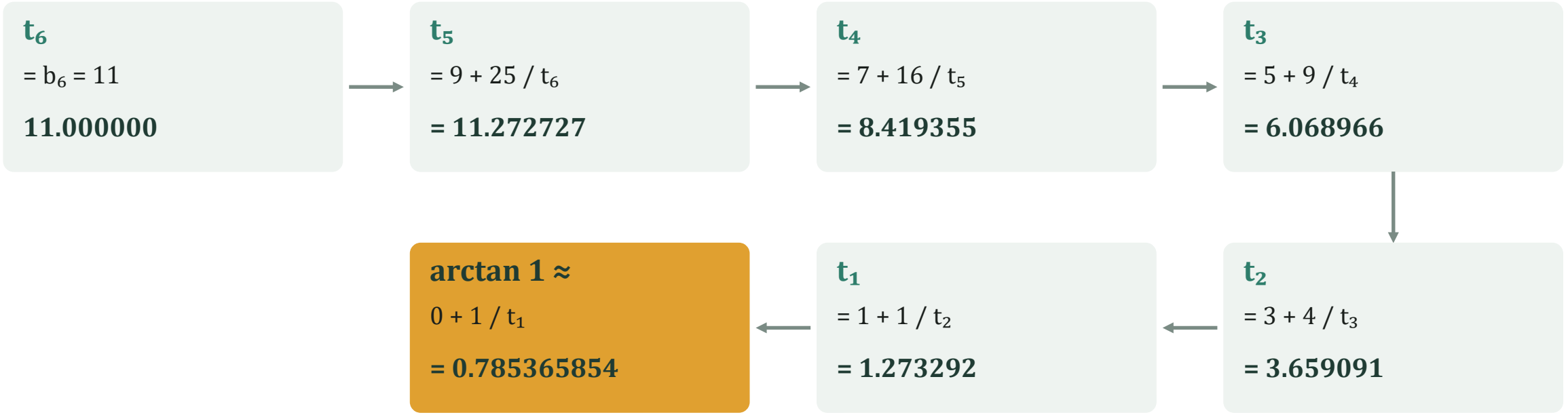
$b_0 = 0,\ a_1 = x,\ b_k = 2k - 1$
$a_k = (k-1)^2 \cdot x^2 \quad (k \geq 2)$

2 **Substitute the numerical data**

$x = 1 \Rightarrow a_k = 1, 1, 4, 9, 16, 25;\ b_k = 1, 3, \ldots, 11$

Truncation: keep 6 levels, i.e. compute the convergent C_6. The table (left, bottom) is the complete input of both algorithms.

CF Example 4 · Step 3: backward evaluation (bottom-up)

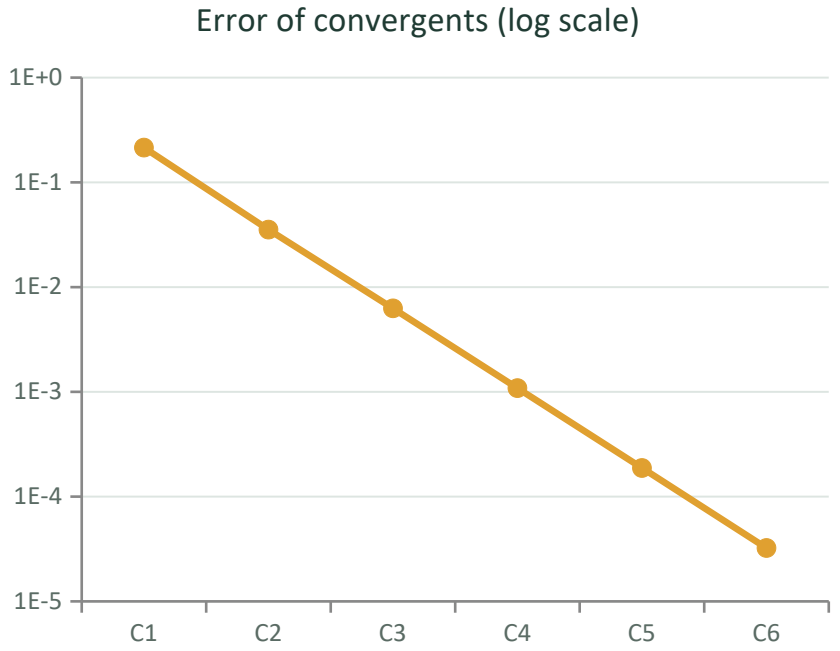


Result: arctan 1 ≈ 0.785365854 Exact: $\pi/4 = 0.785398163397\ldots$
Absolute error: 3.23×10^{-5} obtained with only 6 divisions. Read the snake in the arrow direction: start at the deepest level t_6.

CF Example 4 · Step 4: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	0.7854
1	1	1	1	1	1.000000000	0.2146
2	1	3	3	4	0.750000000	0.0354
3	4	5	19	24	0.791666667	6.27×10^{-3}
4	9	7	160	204	0.784313725	1.08×10^{-3}
5	16	9	1744	2220	0.785585586	1.87×10^{-4}
6	25	11	23184	29520	0.785365854	3.23×10^{-5}



$C_6 = 0.785365854 \equiv$ backward result ✓
≈ 4 correct decimals

CF Example $5 \cdot \sqrt{10} = \sqrt{(3^2 + 1)}$: set up the coefficients

Goal: evaluate √10 with n = 5 levels

$$\sqrt{(a^2+b)} = a + \cfrac{b}{2a + \cfrac{b}{2a + \cfrac{b}{2a + \cdots}}}$$

1 **Read the pattern of a_k and b_k**

$b_0 = a,\ a_k = b,\ b_k = 2a$ (periodic)
from $x = a + b/(a + x)$

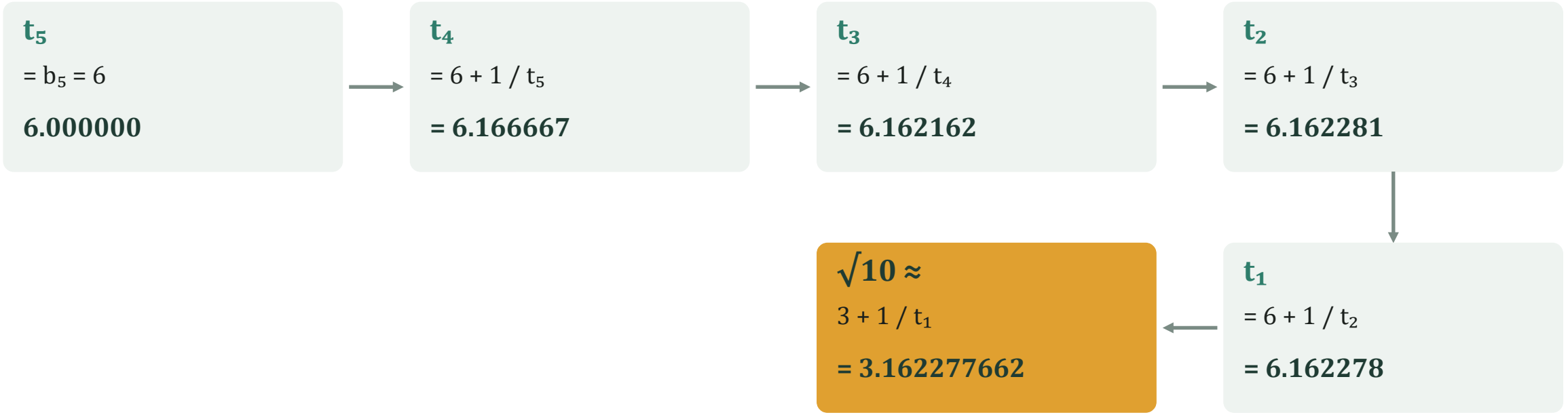
2 **Substitute the numerical data**

$a = 3, b = 1 \Rightarrow b_0 = 3, a_k = 1, b_k = 6$

k	0	1	2	3	4	5
b_k	3	6	6	6	6	6
a_k	—	1	1	1	1	1

Truncation: keep 5 levels, i.e. compute the convergent C_5. The table (left, bottom) is the complete input of both algorithms.

CF Example 5 · Step 3: backward evaluation (bottom-up)

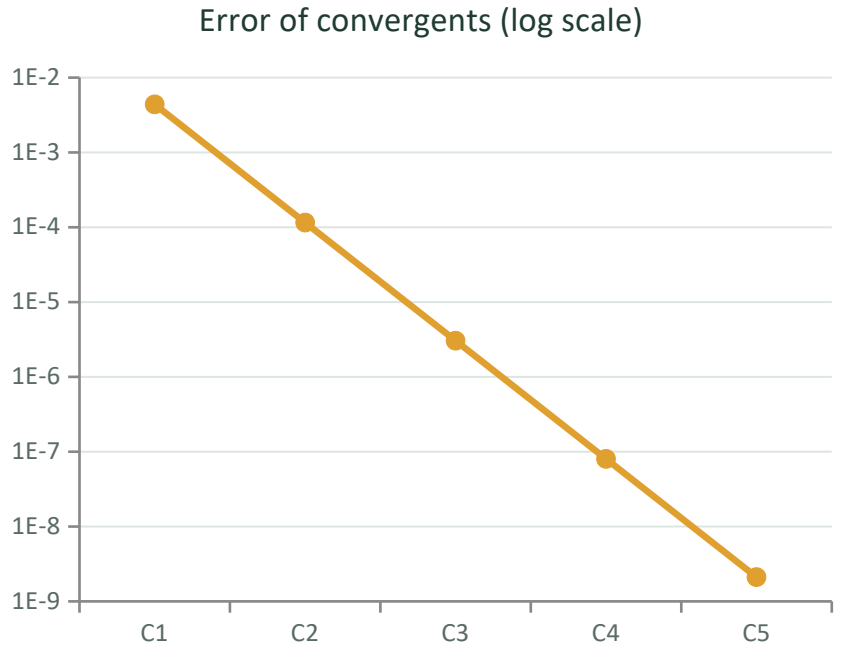


Result: $\sqrt{10} \approx 3.162277662$ Exact: $\sqrt{10} = 3.162277660168\ldots$
Absolute error: 2.11×10^{-9} obtained with only 5 divisions. Read the snake in the arrow direction: start at the deepest level t_5.

CF Example 5 · Step 4: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	3	3	1	3.000000000	0.1623
1	1	6	19	6	3.166666667	4.39×10^{-3}
2	1	6	117	37	3.162162162	1.15×10^{-4}
3	1	6	721	228	3.162280702	3.04×10^{-6}
4	1	6	4443	1405	3.162277580	8.01×10^{-8}
5	1	6	27379	8658	3.162277662	2.11×10^{-9}



$C_5 = 3.162277662 \equiv$ backward result ✓
≈ 8 correct decimals

Part I summary: results of the five continued-fraction examples

Quantity	Levels n	CF value C_n	Exact value	Absolute error
$e = e^1$	5	2.718281829	2.718281828459…	2.77×10^{-10}
$\tan 1$	5	1.557407407	1.557407724654…	3.17×10^{-7}
$\ln 2$	7	0.693152455	0.693147180559…	5.27×10^{-6}
$\arctan 1 = \pi/4$	6	0.785365854	0.785398163397…	3.23×10^{-5}
$\sqrt{10}$	5	3.162277662	3.162277660168…	2.11×10^{-9}

1 Same C_n, two routes

Backward scheme and forward recurrence give identical values — a built-in check.

2 Few levels suffice

e and √10 reach 8–9 correct decimals with only 5 levels.

3 Alternating convergents

C_k oscillate around f when all $a_k > 0$: the true value is bracketed.

Part II · Transcendental equations by Rolle's sequence method

Use the zeros of f’ to count and separate all real roots of f(x) = 0, then refine each one

Rolle's theorem and the key corollary

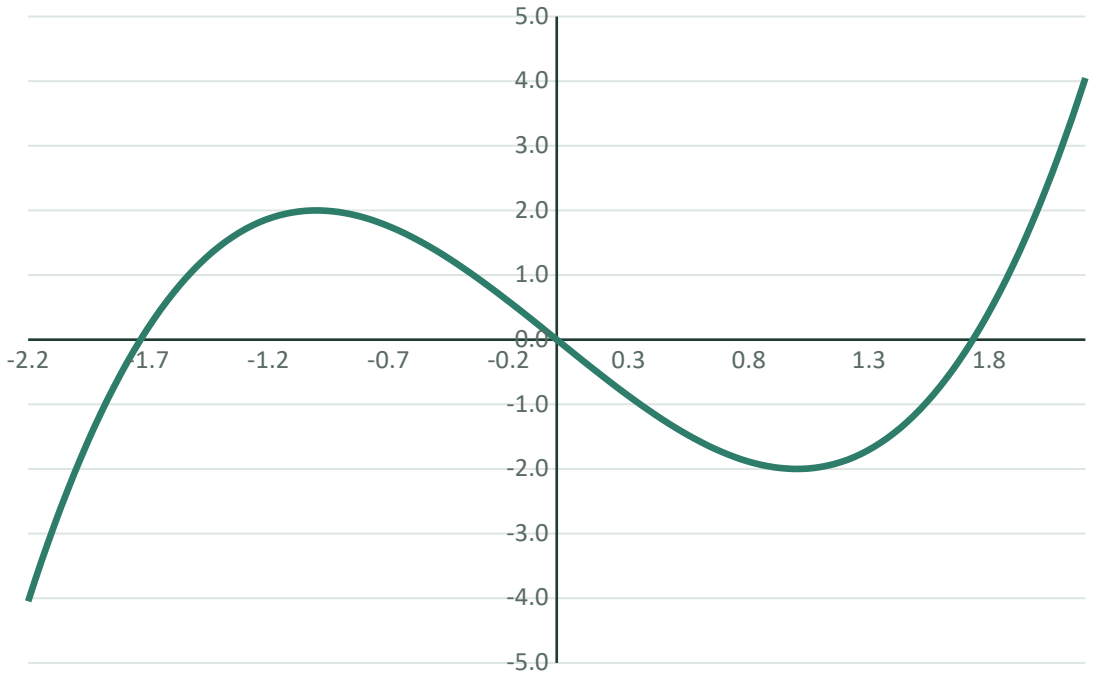
Rolle's theorem

If f is continuous on [a, b], differentiable on (a, b) and f(a) = f(b), then $\exists\, c \in (a, b)$ with $f'(c) = 0$.

Corollary (basis of the method)

- Between two consecutive zeros of f′ there is at most one zero of f.
- There is exactly one if f changes sign there; none otherwise.

f(x) = x³ − 3x : zeros of f′ at ±1 separate the 3 zeros of f



$f'(x) = 3x^2 - 3 = 0 \Rightarrow c = \pm 1$. Signs: $f(-\infty)$ −, $f(-1)$ +, $f(1)$ −, $f(+\infty)$ + ⇒ 3 sign changes ⇒ 3 real roots.

The algorithm of Rolle's sequence method

1 Domain

Fix $D = (\alpha, \beta)$ where f is continuous and differentiable.

2 Derivative

Compute $f'(x)$ and solve $f'(x) = 0$ on D (exactly or numerically).

3 Critical points

Order them: $\alpha < c_1 < c_2 < \ldots < c_k < \beta$.

4 Rolle's sequence

Evaluate $f(\alpha^+), f(c_1), \ldots, f(c_k), f(\beta^-)$ — limits at the ends.

5 Count sign changes

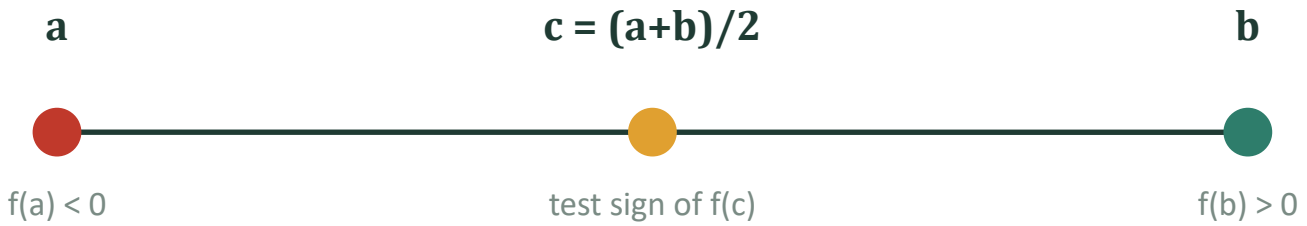
Each sign change ⇒ exactly one root in that interval.

6 Refine

Shrink to a finite bracket [a, b], then bisection (or Newton).

Remarks: if $f(c_i) = 0$, then c_i is a multiple root; number of real roots = number of sign changes in Rolle's sequence.

Refinement step: the bisection method



Keep the half where f changes sign:

- $f(a)\cdot f(c) < 0 \Rightarrow b \leftarrow c$
- otherwise $\Rightarrow a \leftarrow c$
- after n steps: $|\xi - c_n| \le (b-a) / 2^n$
- steps needed for tolerance ε: $n \ge \log_2((b-a)/\varepsilon)$

Reading the tables in the examples

k	iteration number
a, b	current bracket
c	midpoint (new estimate)
f(c)	value at the midpoint, coloured by sign

+ positive **– negative**

7 steps on a bracket of length 1 give an error ≤ 1/128 ≈ 0.0078; we also report the root to 7 decimals (≈ 25 steps).

Rolle Example 1 · $e^x - 3x = 0$: Steps 1–3 (setup)

1 **Domain**

$D = \mathbb{R}$ (f is continuous and differentiable)

2 **Derivative**

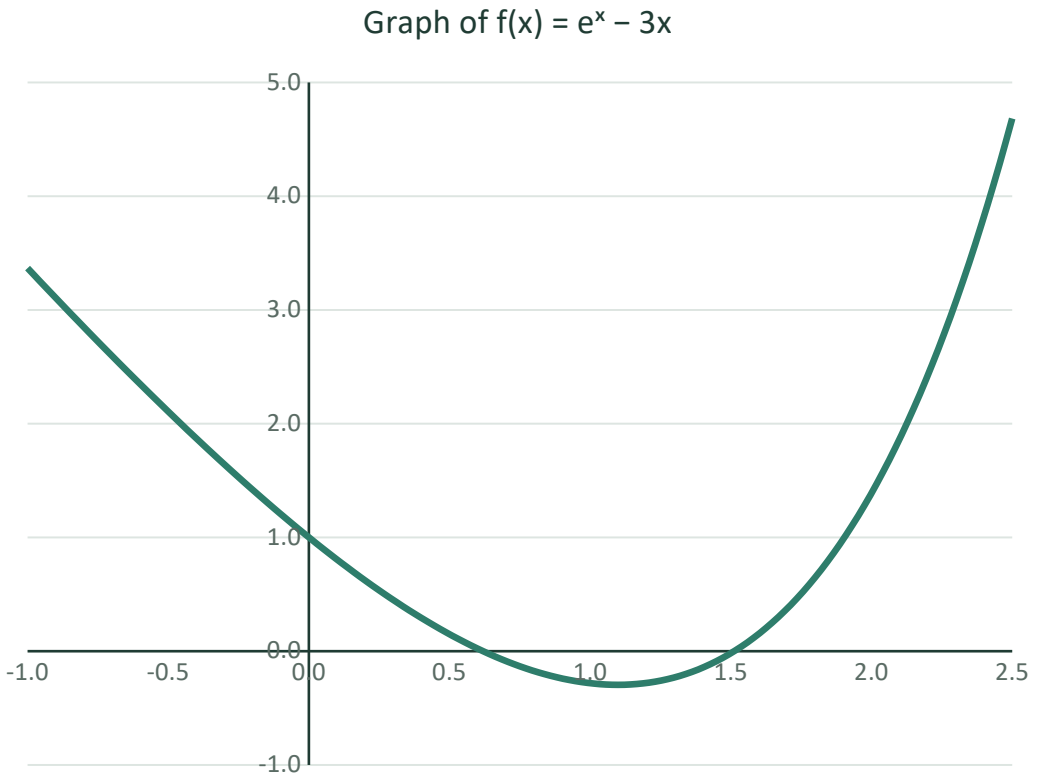
$f'(x) = e^x - 3$

3 **Critical points**

$e^x - 3 = 0 \Rightarrow c_1 = \ln 3 \approx 1.098612$

4 **Limits at the ends**

$x \to -\infty$: $f \to +\infty$; $x \to +\infty$: $f \to +\infty$



f decreases on $(-\infty, \ln 3)$ and increases on $(\ln 3, \infty)$; the minimum $f(\ln 3) = 3 - 3 \ln 3 < 0$.

Rolle Example 1 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = e^x - 3x$

x	$-\infty$	$\ln 3 \approx 1.0986$	$+\infty$
$f(x)$	$+\infty$	-0.295837	$+\infty$
sign	+	−	+



Step 5 · 2 sign changes ⇒ exactly 2 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, \ln 3) \to (0, 1)$

$f(0) = 1 > 0,\ f(1) = e - 3 = -0.2817 < 0$

$\xi_2 \in (\ln 3, +\infty) \to (1, 2)$

$f(1) < 0$ (and $f < 0$ on $[1, \ln 3]$), $f(2) = e^2 - 6 = 1.3891 > 0$

Rolle Example 1 · Step 6: refine by bisection

ξ_1 on [0, 1]

k	a	b	c = (a+b)/2	f(c)
1	0.000000	1.000000	0.500000	**0.148721**
2	0.500000	1.000000	0.750000	**−0.133000**
3	0.500000	0.750000	0.625000	**−0.006754**
4	0.500000	0.625000	0.562500	**0.067555**
5	0.562500	0.625000	0.593750	**0.029516**
6	0.593750	0.625000	0.609375	**0.011156**
7	0.609375	0.625000	0.617188	**0.002145**

$\xi_1 \approx c_7 = 0.617188$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = 0.6190613$

ξ_2 on [1, 2]

k	a	b	c = (a+b)/2	f(c)
1	1.000000	2.000000	1.500000	**−0.018311**
2	1.500000	2.000000	1.750000	**0.504603**
3	1.500000	1.750000	1.625000	**0.203419**
4	1.500000	1.625000	1.562500	**0.083233**
5	1.500000	1.562500	1.531250	**0.030203**
6	1.500000	1.531250	1.515625	**0.005390**
7	1.500000	1.515625	1.507813	**−0.006598**

$\xi_2 \approx c_7 = 1.507813$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 1.5121346$

Rolle Example 2 · ln x – x + 2 = 0: Steps 1–3 (setup)

1 **Domain**

$D = (0, +\infty)$ (because of $\ln x$)

2 **Derivative**

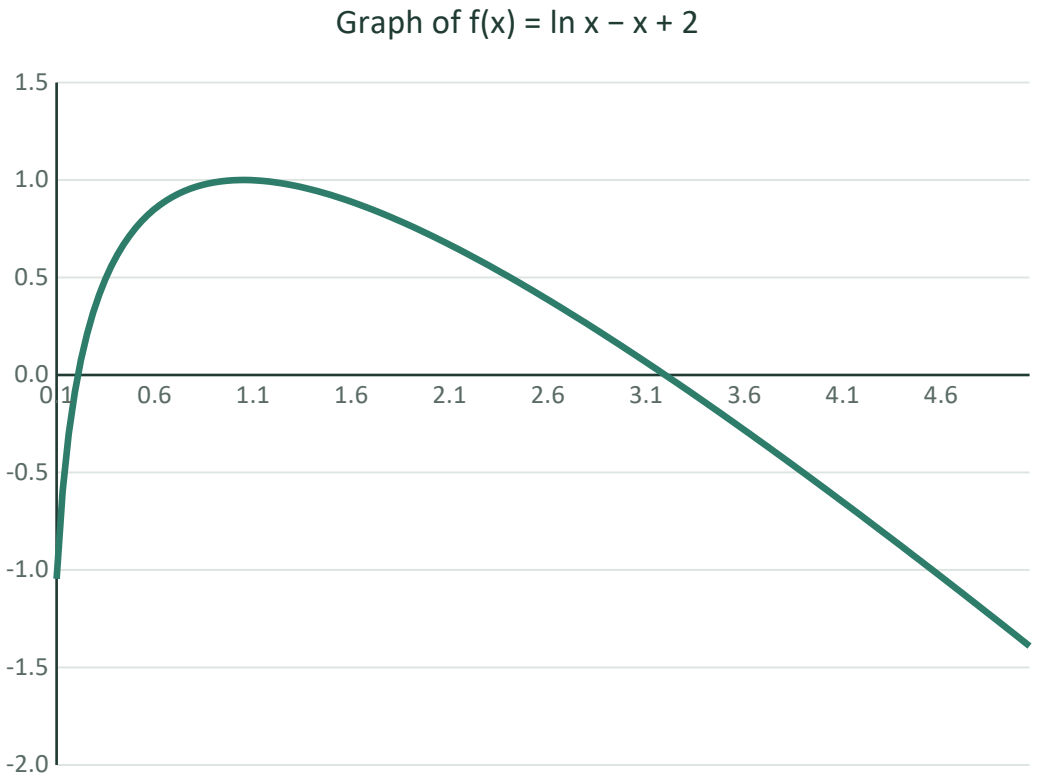
$f'(x) = 1/x - 1 = (1 - x)/x$

3 **Critical points**

$f'(x) = 0 \Rightarrow c_1 = 1$; $f(1) = 0 - 1 + 2 = 1$

4 **Limits at the ends**

$x \to 0^+: f \to -\infty$; $x \to +\infty: f \to -\infty$



f increases on (0, 1) and decreases on (1, ∞); the maximum f(1) = 1 > 0.

Rolle Example 2 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for f(x) = ln x − x + 2

x	0^+	1	$+\infty$
f(x)	$-\infty$	1.000000	$-\infty$
sign	−	+	−



Step 5 · 2 sign changes ⇒ exactly 2 real roots — shrink each interval to a finite bracket

$\xi_1 \in (0, 1) \to (0.1, 0.2)$

$f(0.1) = -0.4026 < 0,\ f(0.2) = 0.1906 > 0$

$\xi_2 \in (1, +\infty) \to (3, 4)$

$f(3) = 0.0986 > 0,\ f(4) = -0.6137 < 0$

Rolle Example 2 · Step 6: refine by bisection

ξ_1 on [0.1, 0.2]

k	a	b	c = (a+b)/2	f(c)
1	0.100000	0.200000	0.150000	**−0.047120**
2	0.150000	0.200000	0.175000	**0.082031**
3	0.150000	0.175000	0.162500	**0.020423**
4	0.150000	0.162500	0.156250	**−0.012548**
5	0.156250	0.162500	0.159375	**0.004130**
6	0.156250	0.159375	0.157813	**−0.004160**
7	0.157813	0.159375	0.158594	**−0.000003**

$\xi_1 \approx c_7 = 0.158594$, $|\text{error}| \le 0.1/2^7 = 0.0008$
Continuing the bisection: $\xi_1 = 0.1585943$

ξ_2 on [3, 4]

k	a	b	c = (a+b)/2	f(c)
1	3.000000	4.000000	3.500000	**−0.247237**
2	3.000000	3.500000	3.250000	**−0.071345**
3	3.000000	3.250000	3.125000	**0.014434**
4	3.125000	3.250000	3.187500	**−0.028263**
5	3.125000	3.187500	3.156250	**−0.006865**
6	3.125000	3.156250	3.140625	**0.003797**
7	3.140625	3.156250	3.148438	**−0.001531**

$\xi_2 \approx c_7 = 3.148438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 3.1461932$

Rolle Example 3 · x – 2 sin x = 0: Steps 1–3 (setup)

1 **Domain and bounding**

$D = \mathbb{R}$; for $|x| > 2$: $|2 \sin x| \le 2 < |x| \Rightarrow$ no roots, so work on $[-2, 2]$

2 **Derivative**

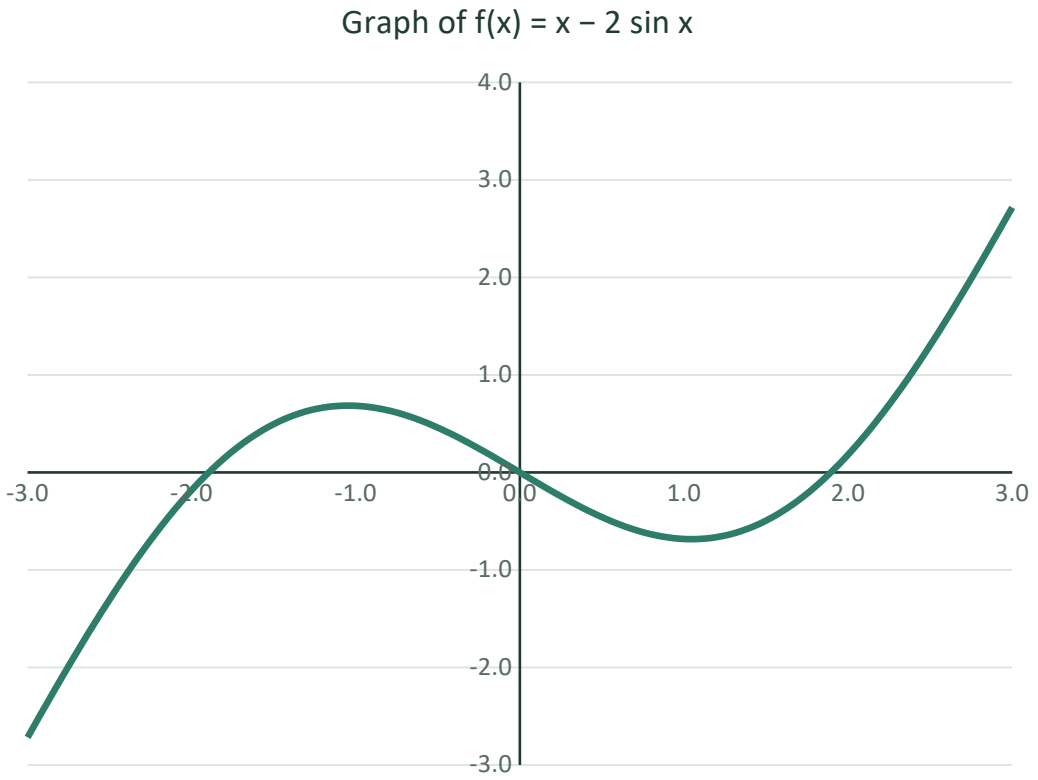
$f'(x) = 1 - 2 \cos x$

3 **Critical points in [–2, 2]**

$\cos x = 1/2 \Rightarrow c_1 = -\pi/3,\ c_2 = \pi/3\ (\pm 1.047198)$

4 **Values at the ends**

$f(-2) = -0.181405,\ \ f(2) = 0.181405$



f is odd: f(−x) = −f(x). So roots come in pairs ±ξ, and x = 0 is a root.

Rolle Example 3 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for f(x) = x − 2 sin x

x	−2	−π/3	π/3	2
f(x)	−0.181405	0.684853	−0.684853	0.181405
sign	−	+	−	+



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-2, -\pi/3) \to (-2, -1.5)$

$f(-2) < 0,\ f(-1.5) = 0.4950 > 0$

$\xi_2 \in (-\pi/3, \pi/3)$

$f(0) = 0 \Rightarrow \xi_2 = 0$ exactly

$\xi_3 \in (\pi/3, 2) \to (1.5, 2)$

$f(1.5) = -0.4950,\ f(2) = 0.1814;\ \xi_1 = -\xi_3$

Rolle Example 3 · Step 6: refine by bisection

ξ_3 on [1.5, 2]

k	a	b	c = (a+b)/2	f(c)
1	1.500000	2.000000	1.750000	**−0.217972**
2	1.750000	2.000000	1.875000	**−0.033172**
3	1.875000	2.000000	1.937500	**0.070471**
4	1.875000	1.937500	1.906250	**0.017728**
5	1.875000	1.906250	1.890625	**−0.007954**
6	1.890625	1.906250	1.898438	**0.004829**
7	1.890625	1.898438	1.894531	**−0.001577**

$\xi_3 \approx c_7 = 1.894531$, $|\text{error}| \le 0.5/2^7 = 0.0039$
Continuing the bisection: $\xi_3 = 1.8954943$

ξ_1 on [−2, −1.5]

k	a	b	c = (a+b)/2	f(c)
1	−2.000000	−1.500000	−1.750000	**0.217972**
2	−2.000000	−1.750000	−1.875000	**0.033172**
3	−2.000000	−1.875000	−1.937500	**−0.070471**
4	−1.937500	−1.875000	−1.906250	**−0.017728**
5	−1.906250	−1.875000	−1.890625	**0.007954**
6	−1.906250	−1.890625	−1.898438	**−0.004829**
7	−1.898438	−1.890625	−1.894531	**0.001577**

$\xi_1 \approx c_7 = -1.894531$, $|\text{error}| \le 0.5/2^7 = 0.0039$
Continuing the bisection: $\xi_1 = -1.8954943$

Symmetry check: the two tables are mirror images (f odd) $\Rightarrow \xi_1 = -\xi_3$. Together with $\xi_2 = 0$ the equation has exactly 3 real roots.

Rolle Example 4 · $x^2 \cdot e^{-x} - 0.1 = 0$: Steps 1–3 (setup)

1 **Domain**

$D = \mathbb{R}$

2 **Derivative**

$f'(x) = 2x \cdot e^{-x} - x^2 \cdot e^{-x} = x(2 - x) \cdot e^{-x}$

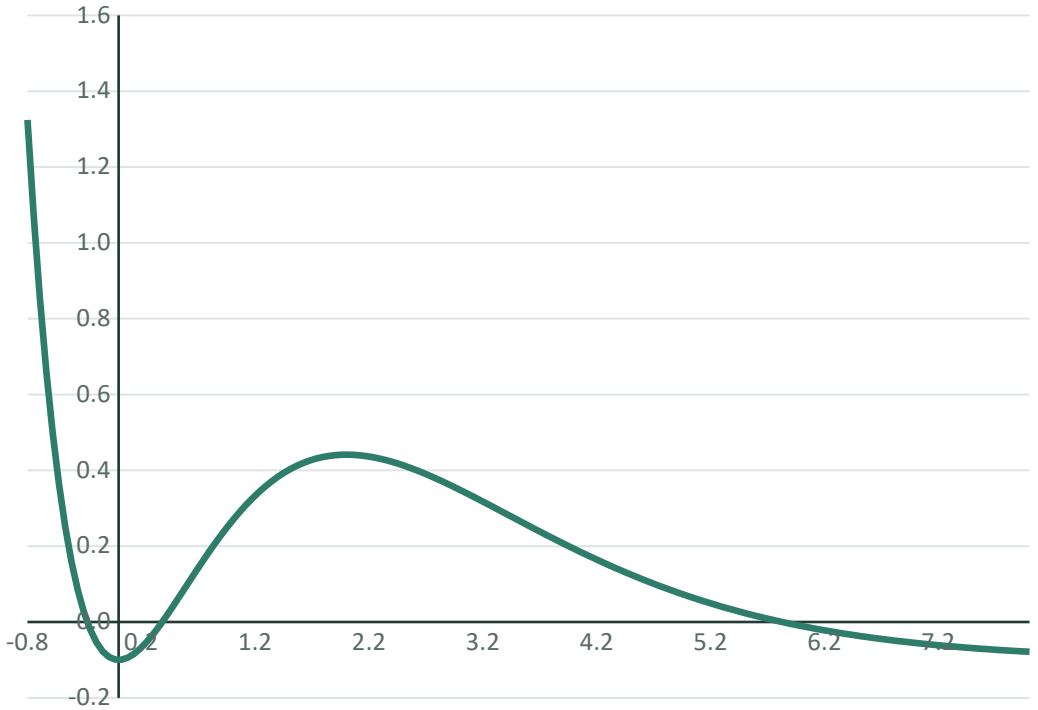
3 **Critical points**

$c_1 = 0$ (min), $c_2 = 2$ (max)

4 **Limits at the ends**

$x \to -\infty$: $f \to +\infty$; $x \to +\infty$: $f \to -0.1$

Graph of $f(x) = x^2 e^{-x} - 0.1$



Rolle values: $f(0) = -0.1$ and $f(2) = 4/e^2 - 0.1 = 0.441341$. Note $f(+\infty) = -0.1$ is a finite limit, still negative.

Rolle Example 4 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = x^2e^{-x} - 0.1$

x	$-\infty$	0	2	$+\infty$
f(x)	$+\infty$	-0.100000	0.441341	-0.1
sign	+	−	+	−



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, 0) \to (-1, 0)$

$f(-1) = e - 0.1 = 2.6183 > 0,\ f(0) = -0.1$

$\xi_2 \in (0, 2) \to (0, 1)$

$f(0) = -0.1 < 0,\ f(1) = 1/e - 0.1 = 0.2679 > 0$

$\xi_3 \in (2, +\infty) \to (5, 6)$

$f(5) = 0.0684 > 0,\ f(6) = -0.0108 < 0$

Rolle Example 4 · Step 6: refine by bisection (part 1)

ξ_1 on [−1, 0]

k	a	b	c = (a+b)/2	f(c)
1	−1.000000	0.000000	−0.500000	**0.312180**
2	−0.500000	0.000000	−0.250000	**−0.019748**
3	−0.500000	−0.250000	−0.375000	**0.104608**
4	−0.375000	−0.250000	−0.312500	**0.033480**
5	−0.312500	−0.250000	−0.281250	**0.004793**
6	−0.281250	−0.250000	−0.265625	**−0.007977**
7	−0.281250	−0.265625	−0.273438	**−0.001719**

$\xi_1 \approx c_7 = -0.273438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = -0.2755303$

ξ_2 on [0, 1]

k	a	b	c = (a+b)/2	f(c)
1	0.000000	1.000000	0.500000	**0.051633**
2	0.000000	0.500000	0.250000	**−0.051325**
3	0.250000	0.500000	0.375000	**−0.003350**
4	0.375000	0.500000	0.437500	**0.023581**
5	0.375000	0.437500	0.406250	**0.009940**
6	0.375000	0.406250	0.390625	**0.003246**
7	0.375000	0.390625	0.382813	**−0.000065**

$\xi_2 \approx c_7 = 0.382813$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 0.3829658$

Rolle Example 4 · Step 6: refine by bisection (part 2)

ξ_3 on [5, 6]

k	a	b	c = (a+b)/2	f(c)
1	5.000000	6.000000	5.500000	0.023625
2	5.500000	6.000000	5.750000	0.005231
3	5.750000	6.000000	5.875000	−0.003053
4	5.750000	5.875000	5.812500	0.001016
5	5.812500	5.875000	5.843750	−0.001037
6	5.812500	5.843750	5.828125	−0.000015
7	5.812500	5.828125	5.820313	0.000499

$\xi_3 \approx c_7 = 5.820313$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_3 = 5.8278978$

All three roots

$\xi_1 = -0.2755303$

$\xi_2 = 0.3829658$

$\xi_3 = 5.8278978$

Check: f(ξ) ≈ 0 for each root, and the number of roots equals the number of sign changes in Rolle's sequence.

Rolle Example $5 \cdot 2^x - x^2 = 0$: Steps 1–3 (setup)

1 Domain

$D = \mathbb{R}$

2 Derivative

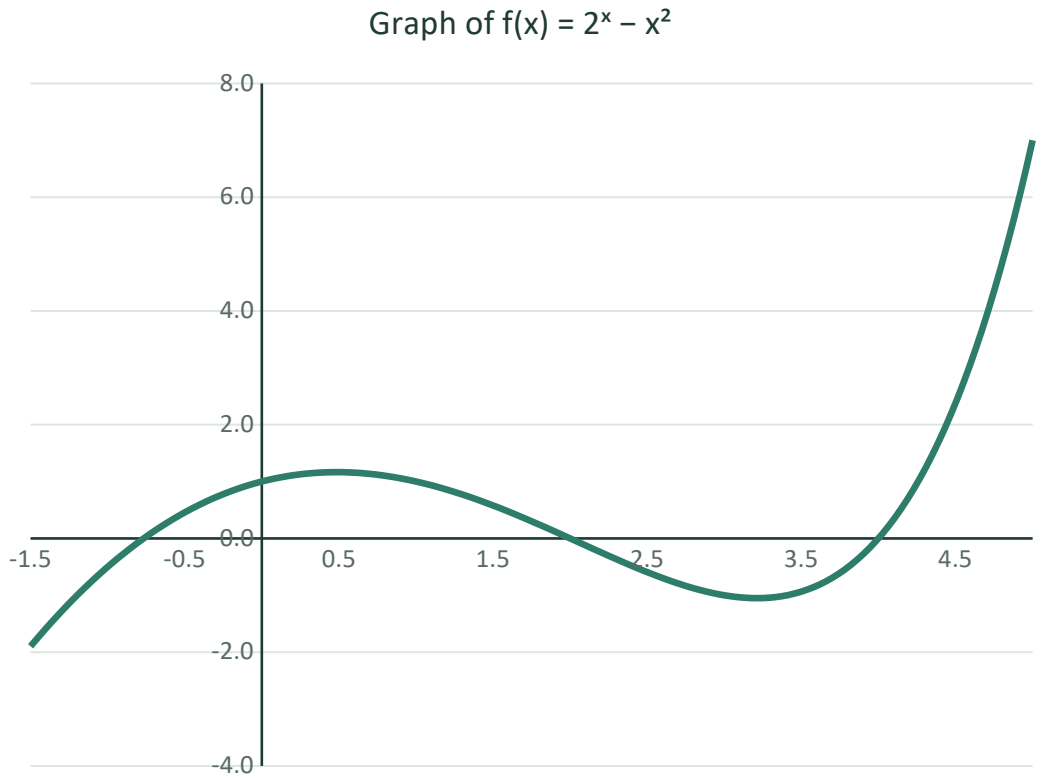
$f'(x) = 2^x \cdot \ln 2 - 2x$

3 Problem

$f'(x) = 0$ is itself transcendental ⇒ apply Rolle's method to f' first

4 Limits at the ends

$x \to -\infty: f \to -\infty; \quad x \to +\infty: f \to +\infty$



Two exact roots can be guessed: $2^2 = 4 = 2^2$ and $2^4 = 16 = 4^2$. Is there a third one? Rolle's sequence will decide.

Rolle Example 5 · Level 1: Rolle's sequence for f′ (via f″)

$f''(x) = 2^x \cdot \ln^2 2 - 2 = 0 \Rightarrow x^* = \log_2(2 / \ln^2 2) \approx 2.057533$ (single zero of f″, solvable exactly)

Rolle's sequence for $f'(x) = 2^x \ln 2 - 2x$

x	$-\infty$	$x^* \approx 2.0575$	$+\infty$
f(x)	$+\infty$	-1.229675	$+\infty$
sign	+	−	+

ξ_1 ξ_2

$c_1 \in (-\infty, x^*) \to (0, 1)$

$f'(0) = \ln 2 = 0.6931 > 0$, $f'(1) = 2 \ln 2 - 2 = -0.6137 < 0$
bisection $\Rightarrow c_1 = 0.485090$

$c_2 \in (x^*, +\infty) \to (3, 4)$

$f'(3) = 8 \ln 2 - 6 = -0.4548 < 0$, $f'(4) = 16 \ln 2 - 8 = 3.0904 > 0$
bisection $\Rightarrow c_2 = 3.212433$

Rolle Example 5 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = 2^x - x^2$

x	$-\infty$	$c_1 \approx 0.4851$	$c_2 \approx 3.2124$	$+\infty$
f(x)	$-\infty$	1.164361	−1.050602	$+\infty$
sign	−	+	−	+



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, c_1) \to (-1, 0)$

$f(-1) = 0.5 - 1 = -0.5 < 0,\ f(0) = 1 > 0$

$\xi_2 \in (c_1, c_2)$

$f(2) = 4 - 4 = 0 \Rightarrow \xi_2 = 2$ exactly

$\xi_3 \in (c_2, +\infty)$

$f(4) = 16 - 16 = 0 \Rightarrow \xi_3 = 4$ exactly

Rolle Example 5 · Step 6: refine ξ_1 by bisection, verify ξ_2 and ξ_3

ξ_1 on [−1, 0]

k	a	b	c = (a+b)/2	f(c)
1	−1.000000	0.000000	−0.500000	**0.457107**
2	−1.000000	−0.500000	−0.750000	**0.032104**
3	−1.000000	−0.750000	−0.875000	**−0.220371**
4	−0.875000	−0.750000	−0.812500	**−0.090762**
5	−0.812500	−0.750000	−0.781250	**−0.028489**
6	−0.781250	−0.750000	−0.765625	**0.002017**
7	−0.781250	−0.765625	−0.773438	**−0.013184**

$\xi_1 \approx c_7 = -0.773438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = -0.7666647$

All three roots

$\xi_1 = -0.7666647$

$\xi_2 = 2 \quad (2^2 = 4 = 2^2)$

$\xi_3 = 4 \quad (2^4 = 16 = 4^2)$

Without Rolle's sequence the negative root is easy to miss; the method also proves there is no fourth root.

Part II summary: roots found by Rolle's sequence method

Equation	# real roots	Roots (7 decimals)
$e^x - 3x = 0$	2	$\xi_1 = 0.6190613;\ \ \xi_2 = 1.5121346$
$\ln x - x + 2 = 0$	2	$\xi_1 = 0.1585943;\ \ \xi_2 = 3.1461932$
$x - 2\sin x = 0$	3	$\xi_1 = -1.8954943;\ \ \xi_2 = 0;\ \ \xi_3 = 1.8954943$
$x^2 \cdot e^{-x} - 0.1 = 0$	3	$\xi_1 = -0.2755303;\ \ \xi_2 = 0.3829658;\ \ \xi_3 = 5.8278978$
$2^x - x^2 = 0$	3	$\xi_1 = -0.7666647;\ \ \xi_2 = 2;\ \ \xi_3 = 4$

1 Count first, compute later

Sign changes give the exact number of real roots before any iteration.

2 Finite limits count too

Ex. 4: $f(+\infty) = -0.1$ is enough to decide the sign at the end.

3 Nested Rolle

If $f' = 0$ is transcendental, apply the method to f' (Ex. 5).

Key takeaways

Continued fractions: few divisions, many digits — evaluate bottom-up or by the A_k, B_k recurrence.

Rolle's sequence: zeros of f′ split the axis; sign changes count and isolate every real root; bisection finishes the job.